

Dynamic Node Selection for Direct Target Tracking Based on Semidefinite Programming

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Abstract—Direct target tracking methods have demonstrated superior tracking accuracy by determining the target positions from the received signals. However, processing raw signals from all sensor arrays in large-scale systems can be computationally inefficient in resource-constrained scenarios. In this paper, we propose a node selection strategy for direct target tracking, where a subset of sensor arrays is adaptively selected to balance tracking performance and computational efficiency. We formulate the node selection problem which aims to minimize the posterior Cramér-Rao lower bound (PCRLB) of the target state estimation error, which jointly integrates both the prior information from the dynamic model and the data information from the received signals. Thereafter, the semidefinite programming technique has been employed to solve the formulated optimization problem. Simulation results reveal that the proposed method exhibits comparable tracking performance to the exhaustive search method.

Index Terms—Distributed sensor arrays, node selection, direct target tracking, semidefinite programming.

I. INTRODUCTION

Target tracking has been a fundamental task in radar and wireless networks, enabling a wide range of applications including surveillance, navigation, and autonomous systems [1]–[3]. Compared with two-step localization and tracking methods, direct position determination (DPD) and direct tracking (DT) methods yield superior tracking accuracy by directly estimating target positions from the received signals without intermediate parameter estimation [4], [5].

Recently, target localization and tracking in distributed sensor arrays have attracted considerable attention owing to their enhanced spatial diversity and improved tracking performance [6], [7]. However, the deployment of a large number of arrays generally increases resource consumption and computational complexity. Especially for DPD and DT in large-scale sensor array networks, processing raw signals collected from all

arrays can be computationally intensive, which would degrade the tracking efficiency in resource-constrained scenarios [8], [9]. Therefore, it is crucial to develop effective node selection strategies to enhance the tracking accuracy while reducing the resource consumption in distributed sensor arrays.

Existing node selection methods are mainly designed for two-step localization and tracking frameworks [10]–[12], where the performance metrics are derived from intermediate parameters, such as time of arrival (TOA) and direction of arrival (DOA), rather than the raw received signals. As a result, the influence of node selection on the signal statistical information is not explicitly characterized [13]. For DPD and DT methods, these two-step-based selection metrics are no longer optimal and even lead to suboptimal tracking performance.

In this work, we propose a node selection strategy for direct target tracking in distributed sensor arrays. The node selection problem is formulated by minimizing the posterior Cramér-Rao lower bound (PCRLB) of the target state estimation error, considering both the information from the target dynamics and the received signals [14]. A semidefinite programming (SDP)-based approach is employed to adaptively select an optimal subset of sensor arrays at each time instant. Based on the selected nodes, direct target tracking is subsequently performed using a particle filter (PF). Numerical results demonstrate the proposed method achieves near-optimal tracking performance compared with the exhaustive search method.

The rest of this paper is organized as follows. Section II introduces the system model, including the signal model and dynamic model. Section III presents the proposed direct tracking framework with the optimal node selection strategy. Numerical simulations are provided in Section IV. Finally, conclusions are drawn in Section V.

II. SYSTEM MODEL

Consider L spatially distributed sensor arrays are employed to intercept signals from the moving target, where each of

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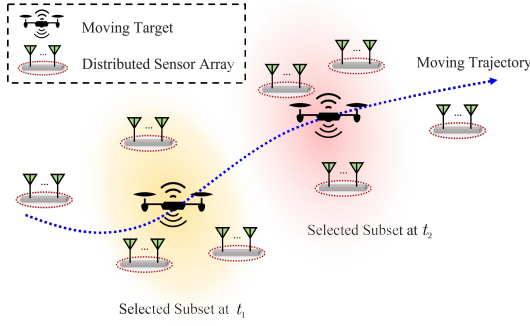


Fig. 1. Tracking scenario with node selection in distributed arrays.

them consists of M elements arranged in a uniform linear array (ULA). The location of the l -th array is denoted by $\mathbf{u}_l = [x_l, y_l]^T$, $l = 1, 2, \dots, L$. The target location at the k -th time instant is represented by $\mathbf{p}_k = [x_k, y_k]^T$, $k = 1, 2, \dots, K$. The interval between two consecutive time instants is denoted by ΔT . The tracking scenario with node selection in distributed arrays is illustrated in Fig. 1.

A. Signal Model

The received signal at the l -th array in the k -th time instant can be expressed as

$$\mathbf{r}_{l,k}(t) = \alpha_{l,k} \mathbf{a}_{l,k} s_k(t - \tau_{l,k}) + \mathbf{n}_{l,k}(t), \quad (1)$$

where $\alpha_{l,k}$ denotes the channel attenuation, $s_k(t)$ is the transmitted signal from the target. The vector $\mathbf{n}_{l,k}(t)$ symbolizes the additive noise at the l -th array, modeled as a complex Gaussian random process with zero mean and covariance matrix $\sigma_n^2 \mathbf{I}_M$. Additionally, $\tau_{l,k}$ and $\mathbf{a}_{l,k}$ denote the propagation delay and the steering vector at the l -th array, respectively.

The received signal in (1) is sampled at a rate of f_s , resulting in N samples collected at each array during the observation interval. The discrete-time representation of the received signal at the k -th time instant can be expressed as

$$\mathbf{r}_{l,k}[n] = \alpha_{l,k} \mathbf{a}_{l,k} s_k[n - \tau_{l,k}] + \mathbf{n}_{l,k}[n], \quad (2)$$

where $n = 0, 1, \dots, N-1$, $s_k[n]$ and $\mathbf{n}_{l,k}[n]$ denote the transmitted signal and noise in temporal domain, respectively. The signal can be transformed into the frequency domain via discrete Fourier transform (DFT),

$$\mathbf{r}_{l,k}[f_n] = \alpha_{l,k} \mathbf{a}_{l,k} s_k[f_n] e^{-j2\pi f_n \tau_{l,k}} + \mathbf{n}_{l,k}[f_n], \quad (3)$$

where $f_n = n f_s / N$ denotes the n -th frequency bin, $s_k[f_n]$ and $\mathbf{n}_{l,k}[f_n]$ represent the n -th frequency component of $s_k(t)$ and $\mathbf{n}_{l,k}(t)$, respectively.

Thereafter, we stacked the signal from all frequency bins into a vector as

$$\mathbf{r}_{l,k} = \alpha_{l,k} \mathbf{A}_{l,k} \mathbf{s}_k + \mathbf{n}_{l,k}, \quad (4)$$

where

$$\begin{aligned} \mathbf{r}_{l,k} &= [\mathbf{r}_{l,k}^T[f_0], \mathbf{r}_{l,k}^T[f_1], \dots, \mathbf{r}_{l,k}^T[f_{N-1}]]^T, \\ \mathbf{s}_k &= [s_k[f_0], s_k[f_1], \dots, s_k[f_{N-1}]]^T, \\ \mathbf{n}_{l,k} &= [\mathbf{n}_{l,k}^T[f_0], \mathbf{n}_{l,k}^T[f_1], \dots, \mathbf{n}_{l,k}^T[f_{N-1}]]^T, \end{aligned} \quad (5)$$

Define

$$\mathbf{D}_{l,k} = \text{diag} \{e^{-j2\pi f_0 \tau_{l,k}}, \dots, e^{-j2\pi f_{N-1} \tau_{l,k}}\}, \quad (6)$$

hence, the steering matrix $\mathbf{A}_{l,k}$ can be expressed as

$$\mathbf{A}_{l,k} = \mathbf{D}_{l,k} \otimes \mathbf{a}_{l,k}, \quad (7)$$

where $\otimes$ represents the Kronecker product.

B. Dynamic Model

The state vector of the moving target at the k -th time instant can be defined as

$$\mathbf{x}_k = [x_k, y_k, v_{x,k}, v_{y,k}]^T, \quad (8)$$

where $v_{x,k}$ and $v_{y,k}$ denote the velocity components along the x and y axes, respectively. The state evolution of the moving target can be modeled using a linear dynamic model as

$$\mathbf{x}_k = \mathbf{F} \mathbf{x}_{k-1} + \mathbf{w}_k, \quad (9)$$

where $\mathbf{F}$ stands for the state transition matrix, and $\mathbf{w}_k$ represents the process noise.

III. DIRECT TARGET TRACKING WITH OPTIMAL NODE SELECTION

A. Posterior Cramér-Rao Lower Bound

We adopt the PCRLB as the performance metric to evaluate the tracking accuracy of the distributed sensor array system. The PCRLB provides a lower bound on the mean square error (MSE) of any unbiased estimator for the target states,

$$\mathbb{E} \{(\hat{\mathbf{x}}_k - \mathbf{x}_k)(\hat{\mathbf{x}}_k - \mathbf{x}_k)^T\} \succeq \mathbf{J}_k^{-1}, \quad (10)$$

where $\hat{\mathbf{x}}_k$ is an unbiased estimate of $\mathbf{x}_k$, and $\mathbf{J}_k$ represents the Bayesian Fisher information matrix (BFIM).

The BFIM $\mathbf{J}_k$ is denoted as

$$\mathbf{J}_k = \mathbb{E}_{\mathbf{x}_k, \mathbf{r}_k} \left\{ -\frac{\partial^2}{\partial \mathbf{x}_k \partial \mathbf{x}_k^T} \ln p(\mathbf{x}_k, \mathbf{r}_k) \right\}, \quad (11)$$

where $\mathbf{r}_k = [\mathbf{r}_{1,k}^T, \mathbf{r}_{2,k}^T, \dots, \mathbf{r}_{L,k}^T]^T$ represents the collection of received signals at the k -th time instant, and $p(\mathbf{x}_k, \mathbf{r}_k)$ denotes the joint probability density function (PDF) of $\mathbf{x}_k$ and $\mathbf{r}_k$. The joint PDF $p(\mathbf{x}_k, \mathbf{r}_k)$ can be factorized as

$$p(\mathbf{x}_k, \mathbf{r}_k) = p(\mathbf{x}_k) p(\mathbf{r}_k | \mathbf{x}_k), \quad (12)$$

where $p(\mathbf{x}_k)$ is the prior PDF of $\mathbf{x}_k$, and $p(\mathbf{r}_k | \mathbf{x}_k)$ denotes the likelihood function of the received signals $\mathbf{r}_k$. Correspondingly, $\mathbf{J}_k$ can be decomposed into two components as

$$\mathbf{J}_k = \mathbf{J}_k^P + \mathbf{J}_k^D, \quad (13)$$

where $\mathbf{J}_k^P$ and $\mathbf{J}_k^D$ represent the prior information matrix and the data information matrix, respectively.

According to [14], the prior term $\mathbf{J}_k^P$ can be obtained by

$$\mathbf{J}_k^P = (\mathbf{F} \mathbf{J}_{k-1}^{-1} \mathbf{F}^T + \mathbf{Q})^{-1}. \quad (14)$$

The data information matrix $\mathbf{J}_k^D$ can be computed as [15]

$$\mathbf{J}_k^D = \frac{2}{\sigma_n^2} \sum_{l=1}^L \text{Re} \left\{ \left(\frac{\partial \mu_{l,k}}{\partial \mathbf{x}_k} \right)^H \frac{\partial \mu_{l,k}}{\partial \mathbf{x}_k} \right\}, \quad (15)$$

where

$$\mu_{l,k} = \alpha_{l,k} \mathbf{A}_{l,k} \mathbf{s}_k. \quad (16)$$

B. Optimal Node Selection by Semidefinite Programming

To reduce the system complexity, we select a subset of the available arrays at each time instant. The selection decision can be represented through a binary selection vector,

$$\mathbf{b}_k = [b_{1,k}, b_{2,k}, \dots, b_{L,k}]^T, \quad (17)$$

where $b_{l,k} \in \{0, 1\}$ indicates whether the l -th array is selected at the k -th time instant ($b_{l,k} = 1$) or not ($b_{l,k} = 0$).

The BFIM $\mathbf{J}_k$ can be rewritten as a function of the selection vector $\mathbf{b}_k$,

$$\mathbf{J}_k(\mathbf{b}_k) = \mathbf{J}_k^P + \mathbf{J}_k^D(\mathbf{b}_k), \quad (18)$$

where $\mathbf{J}_k^D(\mathbf{b}_k)$ stands for the data information matrix corresponding to the selected nodes indicated by the vector $\mathbf{b}_k$. Thus, the optimal node selection problem can be formulated as minimize the PCRLB,

$$\begin{aligned} \min_{\mathbf{b}_k} \quad & \text{tr} \{ \mathbf{J}_k(\mathbf{b}_k)^{-1} \}, \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{b}_k = Q, \\ & b_{l,k} \in \{0, 1\}, \quad l = 1, 2, \dots, L, \end{aligned} \quad (19)$$

where $\text{tr}\{\cdot\}$ denotes the trace of a matrix, Q represents the number of arrays to be selected.

The optimization problem in (19) is NP-hard due to the combinatorial characteristic of the binary selection variables. Specifically, it can be transformed into a semidefinite program, which can be efficiently solved using convex optimization techniques. We first employ a convex relaxation approach to approximate the original problem with a tractable one. The binary constraint on $b_{l,k}$ is relaxed to a continuous interval, i.e., $0 \leq b_{l,k} \leq 1$. Subsequently, an auxiliary variable $\mathbf{Z}$ is introduced to replace the objective function, which satisfies $\mathbf{Z} \succeq \mathbf{J}_k^{-1}(\mathbf{b}_k)$.

Thus, adopting the Schur complement, the optimization problem can be reformulated as

$$\begin{aligned} \min_{\mathbf{b}_k, \mathbf{Z}} \quad & \text{Tr}(\mathbf{Z}) \\ \text{s.t.} \quad & \begin{bmatrix} \mathbf{Z} & \mathbf{I} \\ \mathbf{I} & \mathbf{J}_k(\mathbf{b}_k) \end{bmatrix} \succeq 0, \\ & \mathbf{1}^T \mathbf{b}_k = Q, \\ & 0 \leq b_{l,k} \leq 1, \quad l = 1, 2, \dots, L. \end{aligned} \quad (20)$$

The above problem is a standard SDP and can be solved using convex optimization solvers with interior-point methods [16].

C. Direct Target Tracking with Selected Nodes

The selected nodes at the k -th time instant is defined as $\mathcal{S}_k$, where $\mathcal{S}_k = \{l | b_{l,k} = 1, l = 1, 2, \dots, L\}$. Moreover, the received signals from the selected arrays can be symbolized as $\tilde{\mathbf{r}}_k = \left\{ \mathbf{r}_{l,k}^T \right\}^T$ for $l \in \mathcal{S}_k$.

The PF algorithm consists of two main steps, including prediction and update. In the prediction step, the particles are propagated according to the dynamic model in (9). In the update step, the weights of the particles are updated based on the likelihood of the received signals from the selected arrays,

$$w_k^{(i)} = p(\tilde{\mathbf{r}}_k | \mathbf{x}_k^{(i)}), \quad (21)$$

where $p(\tilde{\mathbf{r}}_k | \mathbf{x}_k^{(i)})$ is the likelihood function of the received signals given the i -th particle state. For each particle, we define the steering vector at the q -th selected array as

$$\mathbf{d}_{q,k} = [d_{q,k}[f_0], d_{q,k}[f_1], \dots, d_{q,k}[f_{N-1}]], \quad (22)$$

where

$$d_{q,k}[f_n] = e^{j2\pi f_n \tau_{q,k}^{(i)}} (\mathbf{a}_{q,k}^{(i)})^H \mathbf{r}_{q,k}[f_n], \quad (23)$$

with $\tau_{q,k}^{(i)}$ and $\mathbf{a}_{q,k}^{(i)}$ denoting the time delay and spatial steering vector at the q -th selected array corresponding to the i -th particle, respectively.

The steering vectors from all selected arrays can be stacked into a matrix as

$$\mathbf{U}_k^{(i)} = [\mathbf{d}_{1,k}^T, \mathbf{d}_{2,k}^T, \dots, \mathbf{d}_{Q,k}^T]^T, \quad (24)$$

Thus, it yields the covariance matrix $\mathbf{C}_k^{(i)}$ as

$$\mathbf{C}_k^{(i)} = \mathbf{U}_k^{(i)} (\mathbf{U}_k^{(i)})^H. \quad (25)$$

According to [5], the likelihood function $p(\tilde{\mathbf{r}}_k | \mathbf{x}_k^{(i)})$ can be calculated by

$$p(\tilde{\mathbf{r}}_k | \mathbf{x}_k^{(i)}) = \lambda_{\max}(\mathbf{C}_k^{(i)}), \quad (26)$$

where $\lambda_{\max}(\cdot)$ denotes the largest eigenvalue of a matrix. The estimated target state at the k -th time instant can be obtained by the weighted average of the particles as

$$\hat{\mathbf{x}}_k = \sum_{i=1}^{N_s} w_k^{(i)} \mathbf{x}_k^{(i)}, \quad (27)$$

where N_s stands for the number of particles.

The overall procedure of direct target tracking with optimal node selection is summarized in Algorithm 1.

IV. SIMULATION RESULTS

In this section, we present simulation results to validate the performance of the proposed optimal node selection method for moving target tracking. Consider a tracking scenario where the target moves in a 2D plane with an initial position of [1000 m, 1000 m] and velocity of [-20 m/s, -20 m/s]. There are $L = 20$ arrays randomly deployed in a square area of 3000 m $\times$ 3000 m, with each array consisting of $M = 5$ antennas. Only $Q = 4$ arrays can be selected for target tracking at each time instant due to resource constraints.

Algorithm 1 Direct Target Tracking with Optimal Node Selection

Input: Candidate sensor arrays $\{\mathbf{u}_l\}_{l=1}^L$, number of selected arrays Q , total tracking time instants K .

Output: Estimated target trajectory.

- 1: Initialize the particles $\{\mathbf{x}_0^{(i)}, w_0^{(i)}\}_{i=1}^{N_s}$, the BFIM $\mathbf{J}_0$, and determine the initial vector $\mathbf{b}_0$ by random selection.
 - 2: **for** $k = 1$ to K **do**
 - 3: Stack the received signals $\tilde{\mathbf{r}}_k$ from the selected arrays.
 - 4: **for** $i = 1$ to N_s **do**
 - 5: Propagate particles using (9).
 - 6: Update particle weights using (21) and (26).
 - 7: **end for**
 - 8: Estimate target state using (27).
 - 9: Update the BFIM $\mathbf{J}_k$ based on the estimated state.
 - 10: Solve the optimization problem in (20) to obtain the optimal selection vector for the next time instant.
 - 11: **end for**
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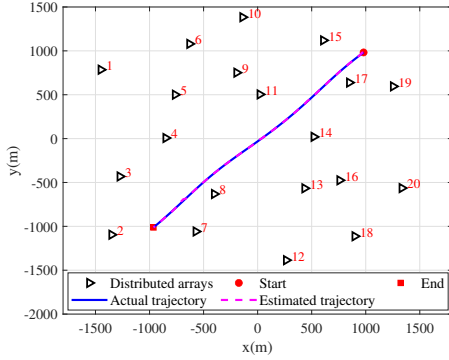


Fig. 2. Tracking results.

The total number of the tracking time instants is set to $K = 100$, and the time interval between two consecutive time instants is $\Delta T = 1$ s. The target transmits a binary phase shift keying (BPSK) signal continuously during the tracking process. Furthermore, the reference SNR at the distance of 1000 m is set to 0 dB, and the signal attenuation follows the free-space path loss model.

Fig. 2 illustrates the tracking results of the proposed method over the entire tracking duration. It can be observed that the estimated trajectory closely follows the actual trajectory throughout the tracking process, demonstrating the proposed node selection strategy is capable of maintaining stable and accurate tracking performance over time.

For a quantitative comparison, we conduct 100 independent Monte Carlo simulations to evaluate the tracking performance of different node selection methods versus the number of selected arrays. Fig. 3 illustrates the average root mean square error (RMSE) of the target position estimation as a function of the number of selected arrays Q over the entire trajectory, while Fig. 4 shows the corresponding average PCRLB. Since the PCRLB provides a theoretical lower bound on the mean-square tracking error, its square root can be regarded as the

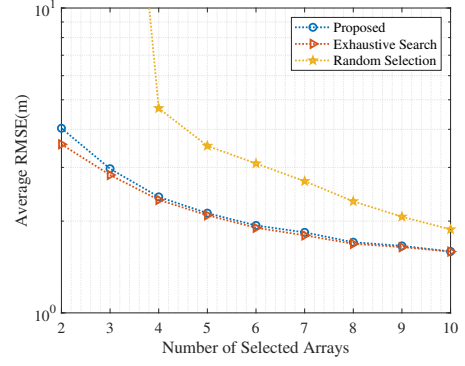


Fig. 3. Average RMSE v.s. number of selected arrays.

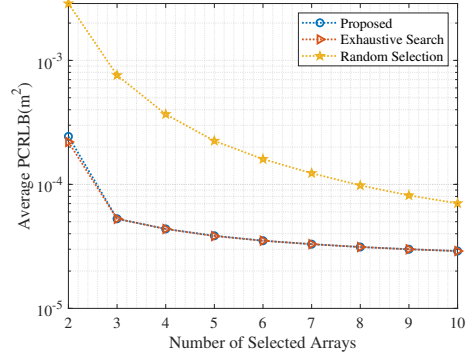


Fig. 4. Average PCRLB v.s. number of selected arrays.

corresponding lower bound for the RMSE. Therefore, Fig. 4 complements Fig. 3 by showing that the RMSE improvement of the proposed method is consistent with its lower theoretical error bound. It can be observed that both the RMSE and PCRLB decrease as the number of selected arrays increases for all methods. Moreover, the proposed method consistently outperforms the random selection approach across all values of Q . The results in Fig. 3 indicate that the proposed method exhibits performance comparable to that of exhaustive search, demonstrating its effectiveness in selecting optimal nodes.

V. CONCLUSION

This paper investigated the problem of optimal node selection for direct target tracking in a distributed sensor array network. We derived the PCRLB for the moving target's trajectory estimation based on the received signals directly, and formulated the optimal node selection problem to minimize the PCRLB at each time instant. To solve the NP-hard problem, we proposed a convex relaxation approach to transform it into a standard SDP, which can be solved using convex optimization techniques. Simulation results demonstrated that the proposed method can achieve near-optimal node selection performance, demonstrating its potential for target tracking applications in resource-constrained sensing networks. Future work will focus on extending the proposed framework to more practical scenarios with time-varying communication constraints and multi-target tracking scenarios.

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